

Stripe Subscription Retention Analysis: Customer Churn & Pricing Tier Analysis | 2023-2024

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Date: May 2026

Context:

This analysis examines how Stripe's pricing tiers, Starter, Growth, and Enterprise, affect customer retention and lifecycle behavior. The goal was to understand subscription-level churn patterns and surface actionable insights to refine pricing and retention strategy. The analysis uses a synthetic Stripe-style dataset of 1,000 customers generated in Python, covering subscription activity from January 2023 to June 2024, with fields including MRR, churn date, payment failures, and feature usage score.

Key Findings:

- Starter tier has 40.5% churn rate, more than 4x the Enterprise rate of 9.5%, suggesting significant price sensitivity or product-fit gaps at the entry-level
- Despite Starter having the highest churn rate, Growth tier represents the largest revenue risk with USD 6,162 MRR at risk, 37% more than Starter's USD 5,655 due to its higher per-customer price.
- Churned customers averaged 2.38 payment failures versus 0.63 for active customers, nearly 4x higher, with 93% of churned customers experiencing at least one payment failure, making it a strong leading indicator of churn.
- Starter and Growth customers begin churning before day 50, while Enterprise customers remain stable for the first 130 days, pointing to an onboarding and early engagement gap in lower tiers.

Recommendations:

- Prioritize retention efforts on the Growth tier first, despite having a lower churn rate than Starter, it represents the largest revenue risk at USD 6,162 MRR at risk. A dedicated account check-in program at the 60 and 90 day mark could identify at-risk customers before they cancel.
- Implemented a structured 30-day onboarding program for Starter and Growth customers, including guided setup, feature walkthroughs and proactive check-ins to address the early churn signal that begins before day 50 for both tiers.
- Build a payment failure intervention system that triggers automated outreach at the first failure, smart retries, dunning emails, and prompts to update payment methods, given that 93% of churned customers experienced at least one payment failure.

Caveats:

- This analysis uses synthetic data generated to mimic Stripe's subscription structure. Real Stripe data may reveal additional churn drivers, different tier distributions, or behavioural patterns not captured here, meaning findings should be validated against actual production data before informing strategy.

- Newer cohorts such as June 2024 show higher retention rates, but this is likely due to insufficient time to churn rather than genuine improvement. These cohorts should be excluded from strategic decisions until at least 6 months of data is available.
- Payment failures and churn show a strong correlation in this dataset, but correlation does not imply causation. Other factors — such as competitor pricing, product dissatisfaction, or seasonal budget cycles — may be driving churn independently of payment failures.